

Bank Customer Churn Analysis

SQL Server + Power BI Portfolio Report

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Dataset: 10,000 bank customers

Overall churn: 20.38%

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Executive Summary

This project examines bank customer churn using SQL Server for validation and segmentation and Power BI for visualization. The SQL work confirms the customer population and the uniqueness of CustomerId, calculates the overall churn rate, and breaks churn down by activity status, complaints, geography, card type, and satisfaction score, finishing with a combined risk-factor query. The Power BI dashboard then presents these findings as KPI cards and segment-level visuals for a non-technical audience.

Business Problem

Retention budget is limited, and spending it evenly across all customers wastes money on customers who were never going to leave. The bank needs to know which segments churn at unusually high rates, and under what conditions, so retention efforts can be targeted rather than applied broadly.

Project Objectives

- Validate the imported customer population and customer identifiers.
- Calculate the overall churn rate and the number of customers who exited.
- Compare churn across customer and banking segments.
- Identify high-risk combinations using geography, complaints, and activity status.
- Present the results through a Power BI churn dashboard.

Dataset Overview

The dataset contains 10,000 customers with demographic and banking attributes, including geography, gender, age, credit score, account balance, tenure, number of products, card type, active-member status, complaint status, satisfaction score, and points earned. The Exited field is the target variable, indicating whether the customer left the bank.

SQL Analysis

The supplied SQL was reviewed in full. It contains customer-count validation, a duplicate-key check, KPI calculations, grouped churn analysis across five dimensions, and a final multi-dimensional risk profile.

EDA & Data Cleaning

Data quality checks in the supplied SQL are structural rather than value-level: a row-count validation and a CustomerId duplicate check. No value-level cleaning transformations, such as TRIM, REPLACE, NULL

replacement, date conversion, or removal of duplicate rows, are present in the supplied SQL, so none are claimed here.

Important SQL Techniques

The project demonstrates COUNT, SUM, CAST, GROUP BY, HAVING, ORDER BY, percentage calculations, and multi-column segmentation. The supplied SQL does not contain joins, CTEs, CASE statements, window functions, RANK / DENSE_RANK / ROW_NUMBER, PARTITION BY, running totals, or SQL views, and none are claimed here.

Selected SQL Example

Overall churn KPI

```
SELECT COUNT(*) AS Total_Customers, SUM(CAST(Exited AS INT)) AS Customers_Who_Left,
CAST(SUM(CAST(Exited AS INT)) * 100.0 / COUNT(*) AS DECIMAL(5,2)) AS Churn_Percent
FROM dbo.Bank_Customer_Churn;
```

This query calculates the customer population, the number of customers who exited, and the overall churn percentage.

High-risk profile analysis

```
SELECT Geography, Complain, IsActiveMember, COUNT(*) AS Total, SUM(CAST(Exited AS INT)) AS
Customers_Left, CAST(SUM(CAST(Exited AS INT)) * 100.0 / COUNT(*) AS DECIMAL(5,2)) AS
Churn_Percent
FROM dbo.Bank_Customer_Churn
GROUP BY Geography, Complain, IsActiveMember
ORDER BY Churn_Percent DESC;
```

This query segments customers by geography, complaint status, and activity status to expose the highest-risk combinations in a single result set.

Power BI Development

The supplied dashboard image supports documentation of the KPI cards and the visible churn visualizations. Specific Power Query steps, table relationships, and DAX formulas were not supplied with the project materials, so they are not claimed as implemented in this report.

Dashboard KPIs

KPI	Value
Total Customers	10,000
Active Customers	7,962
Customers Left	2,038
Churn Rate	20.38%
Average Credit Score	650.53

Key Business Insights

- Germany records the highest geographic churn rate at 32.4%, ahead of Spain at 16.7% and France at 16.2%.
- Customers aged 50–60 have the highest visible age-group churn rate at 56.0%.
- Churn rises sharply for customers with three products (82.7%), and the dashboard shows 100.0% churn for customers with four products. These are the smallest customer groups in the dataset, so the percentage is more volatile and should be treated as a signal to investigate rather than a confirmed trend.
- Female customers show a 25.1% churn rate versus 16.5% for male customers in the dashboard.
- Overall churn is 20.38%, with 2,038 customers shown as having exited.

Business Recommendations

- Prioritize retention analysis for customers in Germany, where churn is roughly double the next-highest market.
- Investigate the 50–60 age segment for service, product, or relationship factors linked to churn.
- Review customers holding three or four products to understand whether product complexity, dissatisfaction, or account structure is associated with exits, validating findings against segment size before acting.
- Examine the reasons behind the higher observed female churn rate before designing targeted retention actions.
- Combine complaint, activity, and geography indicators in future analysis to build a more detailed customer-risk framework, and, where available, pair these findings with revenue-per-customer data to prioritize retention spend by financial impact rather than churn rate alone.

Limitations

- The dataset is a single snapshot rather than a time series, so churn drivers cannot be confirmed as causal, only associative.
- The three- and four-product segments contain far fewer customers than the one- and two-product segments, which can make their churn percentages appear more extreme than the underlying sample size supports.
- Power Query steps, data-model relationships, and DAX measures were not supplied with the project materials, so the Power BI implementation details are documented from the dashboard image only.

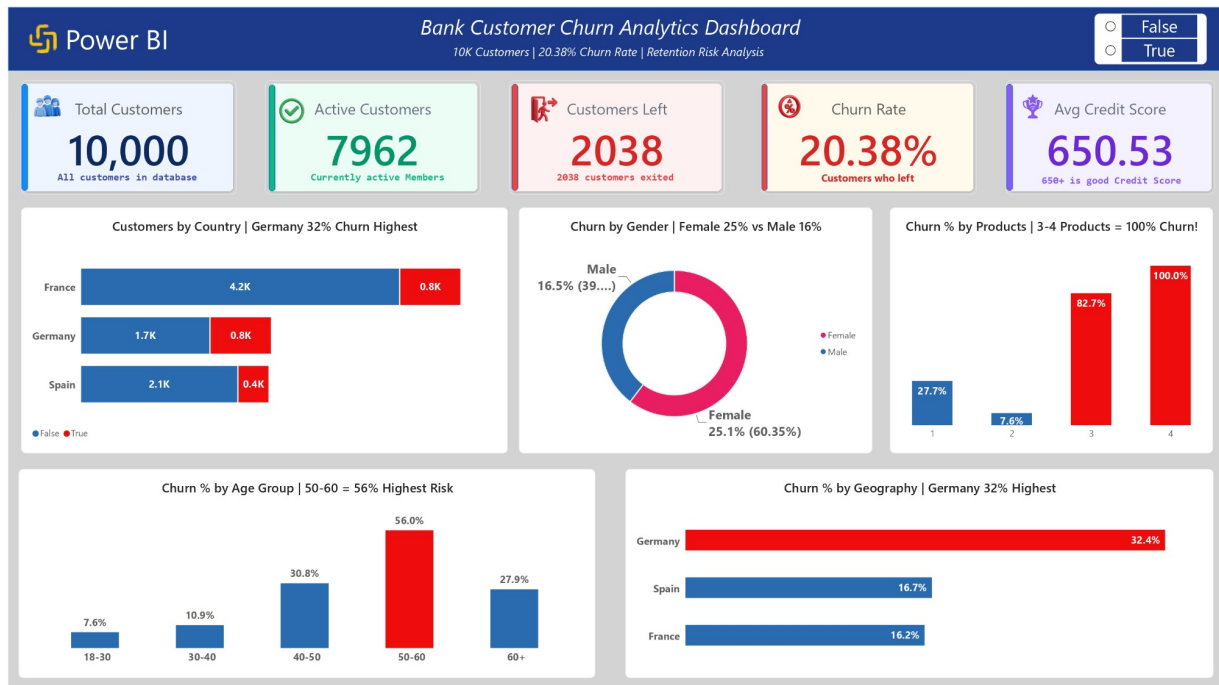
Results

The project produces a focused churn analysis from SQL Server and a visual Power BI summary. Churn risk is most concentrated in Germany, the 50–60 age group, and customers holding three or more products, giving the bank a specific, data-backed starting point for retention efforts rather than a blanket approach across the full customer base.

Conclusion

This project demonstrates a complete, honestly-scoped portfolio workflow: SQL Server is used to validate and segment the data, and Power BI is used to turn those findings into a dashboard a non-technical stakeholder can explore directly. Where project materials were incomplete, such as the absence of a supplied .pbix file, that gap is disclosed rather than papered over, so every claim in this report can be traced back to the supplied SQL file or dashboard image.

Dashboard Screenshot



Source: Power BI dashboard image supplied for this project.

Author

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